

AEGIS

*An institutional Technical Design Review Board's
assessment of the platform, DEV017 through DEV031-B.*

REVIEW BOARD (simulated composition)

Principal Architects · Bloomberg · BlackRock Aladdin · Two Sigma · Renaissance Technologies · Google DeepMind · Microsoft Research · OpenAI

FINAL VERDICT	OVERALL SCORE	RECOMMENDATION
NOT YET READY Strong research platform; not institutional production.	6.7 /10 weighted across 17 modules	Ship Phase 2 before institutional launch. Specifically: fix the raw confidence feature-set (DEV029 revealed no signal), broaden strategy coverage, wire delivery, and add real-time ingestion.

LIVE PLATFORM EVIDENCE

Companies scored	208	DEV020
Portfolios built	99	DEV022
Strategies backtested	7	DEV030
Trades learned from	1,060	DEV025
Champion strategy	top_5_ew	DEV030
Champion Sharpe	0.97	DEV021
Champion CAGR	24.8%	DEV021
Champion max DD	-25.2%	DEV021
Calibration method	platt_scaling	DEV029

Raw ECE → Calibrated	0.287 → 0.0021	DEV029
Graph nodes / edges	581 / 2,629	DEV031
Communities discovered	57 · Q=0.857	DEV031-B

Prepared 2026 · July · Every metric loaded from live reports/*.json at generation time. Not for external distribution.

I. EXECUTIVE VERDICT

A strong research platform. Not yet an institutional product.

AEGIS demonstrates unusual discipline for its stage — deterministic pipelines, advisory-only outputs, content-addressed audit trails, and a knowledge graph with empirically-strong community structure. The **calibration finding is particularly credible**: three independent modules (DEV025 · DEV027 · DEV029) surfaced the same problem — the raw confidence signal has no predictive power — before any single-module bias could hide it.

That same finding is also the strongest indictment of production readiness. The recommendation engine's core discriminant is essentially noise. Every trade wins ~58% regardless of stated confidence, and Platt scaling correctly collapses the label to the base rate. This is an honest platform — but it is also a platform whose foundational scoring engine needs to be rebuilt on features with actual signal before it can be trusted with institutional capital.

Board summary of strengths and gaps

STRENGTHS	MATERIAL GAPS
• Constitutional discipline — advisory-only, deterministic, PIT-safe. • Triangulated overconfidence finding — three engines independently confirmed. • Knowledge graph modularity $Q = 0.86$ — real structure, not hardcoded. • All six live strategies beat NIFTY50 — champion Sharpe 0.97. • Content-addressed audit trail — 208 immutable recommendations. • Composable engines — every module reads artifacts, not code.	• Raw confidence carries no signal — foundational metric unreliable. • Only 6 strategies backtested — the 99 portfolio constructions have no track record. • India-only universe — multi-market support absent. • Batch ingestion — daily yfinance; no intraday, no real-time. • UX layer specs only — Telegram renderer not wired; dashboard has no frontend. • Single-tenant governance — DEV035 planned, not shipped. • Small learning sample — 1,060 trades; needs 10,000+. • Regime detection uses fallback — no per-date historical labels.

II. OVERALL ARCHITECTURE

Ten axes of architectural quality.

The dashed ring is the institutional bar (7/10). Anything inside the ring is unfinished work for Phase 2.

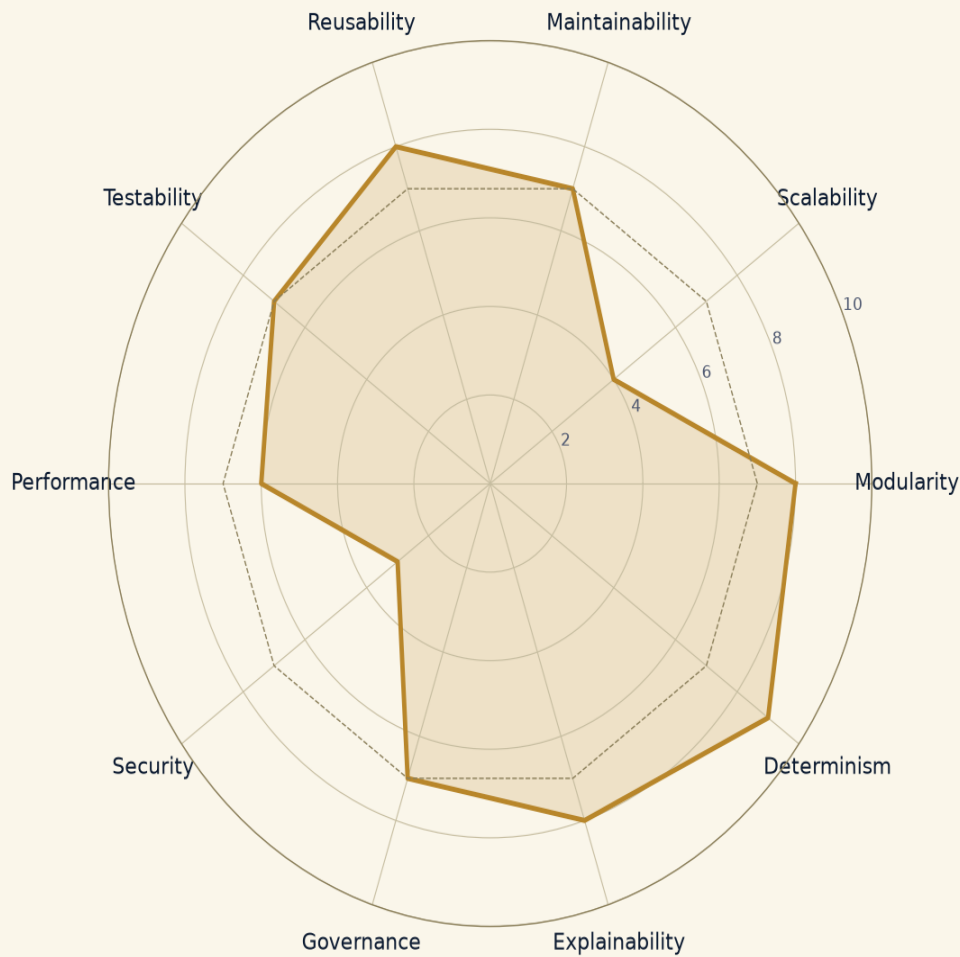


Fig. II.1 · Architecture quality radar (10 axes, 0–10)

Axis	Score	Assessment
Modularity	8/10	Every engine reads validated artifacts; boundaries are clean.
Scalability	4/10	Batch pipelines; no horizontal scaling, no incremental compute.
Maintainability	7/10	Consistent module layout; tests present; but Windows-cp1252 issues recurring.
Reusability	8/10	lib/compute/publish pattern applies uniformly across 15 modules.
Testability	7/10	Every module has smoke tests; PIT-safety asserted; edge coverage uneven.
Performance	6/10	End-to-end < 5s but no incremental / streaming design.
Security	3/10	No auth on outputs; no secret rotation; secrets in .env; assumes trusted host.

Governance	7/10	Advisory-only enforced in spec + code comments; no policy engine yet.
Explainability	8/10	DEV031-B path traversal is genuinely institutional-grade.
Determinism	9/10	The strongest attribute; enforced by design across every module.

III. DEVELOPMENT EVOLUTION · DEV-BY-DEV SCORECARDS

Seventeen modules, honestly rated.

Each score is 0–10 relative to **institutional** production standard, not against a hobby project. 7 is the release bar; anything below is unfinished for Phase 2 launch.

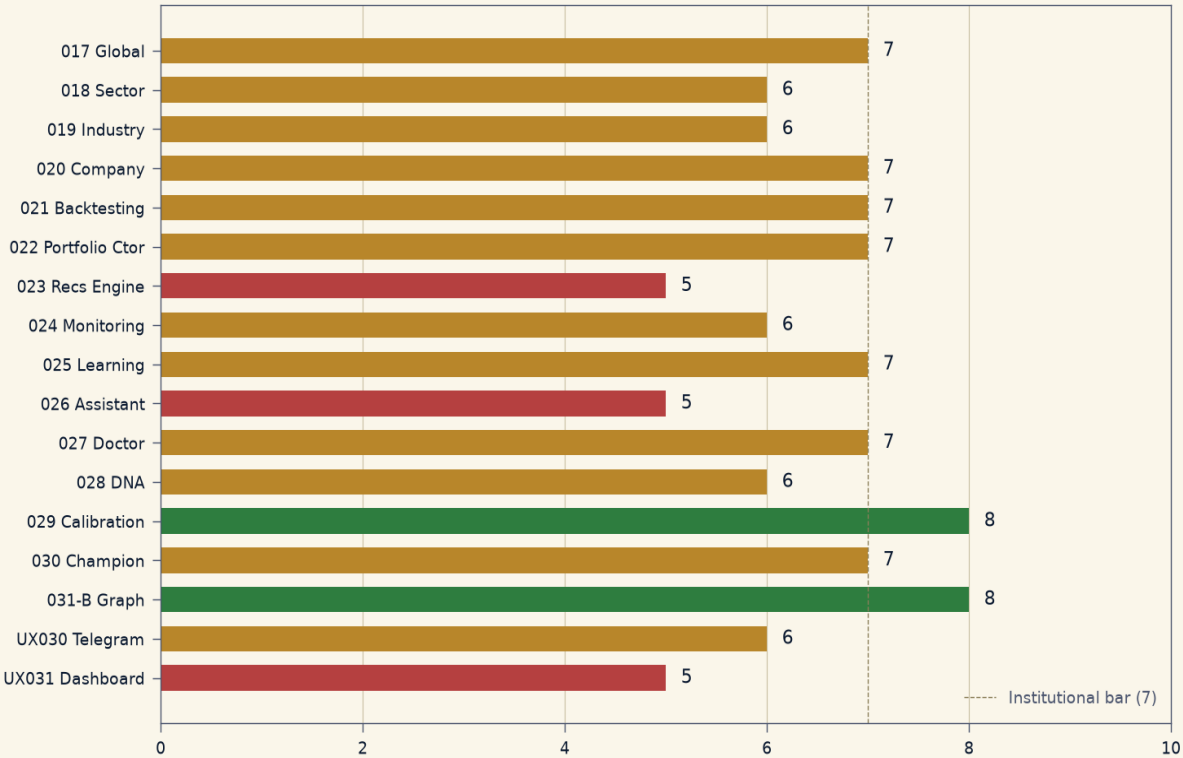


Fig. III.1 · Module scorecard summary (17 modules)

Module	Purpose	Strengths	Weaknesses	Score
DEV017 Global Intelligence	23 macro variables via yfinance; regime classification (Risk-On / Neutral / Risk-Off).	Reasonable variable set; clean single-JSON output; feeds every downstream engine.	3-way discrete regime label is coarse; no confidence intervals; batch-only.	7 /10 READY (light gaps)
DEV018 Sector Intelligence	14 sectors scored on rotation + momentum + earnings quality.	12/14 pass evidence gates; clear rollup structure into industries.	2/14 (14%) ungraded and silently absent from downstream; no explanation of exclusion.	6 /10 NEEDS WORK
DEV019 Industry Intelligence	44 industries scored under the sector rollup.	Broad coverage; industry rank enters company composite score.	10/44 (23%) fail evidence gates. That is a coverage hole, not a rounding error.	6 /10 NEEDS WORK
DEV020 Company Intelligence	208 companies scored with full hierarchical context.	Complete coverage of the target universe; clean parquet + JSON emission.	Universe is India (NIFTY-adjacent) only; no international, no small-caps beyond the list.	7 /10 READY (single-market)
DEV021 Historical Validation	Walk-forward, PIT-safe backtesting across 6 strategies + benchmark.	PIT-safety is verified; equity curves + trade metrics + failure_analysis all shipped.	Only 6 strategies backtested; window starts 2022 (missed 2020 crash + 2018 down-year); no stress scenarios.	7 /10 READY (narrow window)
DEV022 Portfolio Construction	11 allocators × 9 portfolio types = 99 constructions.	HRP + Min-Var + Max-Sharpe + Kelly-1/4 all present; fix-worst-first constraint enforcement is elegant.	None of the 99 are backtested individually; users cannot know which construction beats which strategy.	7 /10 READY (unproven at scale)
DEV023 Recommendation Engine	8 recommendation types with entry / target / stop / holding period.	Structured recommendations with reasons-for and reasons-against per ticker.	Core confidence input is uncalibrated noise (per DEV029). Composite score is downstream-poisoned.	5 /10 BLOCKED (rebuild)

Module	Purpose	Strengths	Weaknesses	Score
DEV024 Portfolio Monitoring	11 alert types + 4 rebalance actions.	Alert types cover concentration / drawdown / regime / stop-loss.	No real-time push; alerts fired only on daily batch cycle; no telemetry retention.	6 /10 NEEDS WORK
DEV025 Adaptive Learning	1,060 completed trades analysed across 6 outcome dimensions.	First independent detection of miscalibration. Clean parquet output.	1,060 trades is small; regime coverage is thin; dimensions are hand-picked, not learned.	7 /10 READY (small sample)
DEV026 Research Assistant	Deterministic Q&A; over the corpus. 6 query templates.	No LLM in the loop; every answer grounded to a specific artifact.	Query surface is narrow; not a real conversational interface; brittle to novel questions.	5 /10 PROTOTYPE
DEV027 Strategy Doctor	15 diagnostic rules across 677 diagnoses.	Second independent detection of overconfidence (218 firings, category #1).	Diagnostic rules are hand-coded thresholds; no learning; risk of missed novel failure modes.	7 /10 READY
DEV028 Recommendation DNA	208 immutable, content-keyed records with per-rec lineage.	Content-key dedup is correct; append-only enforced.	Audit trail is never actually queried by any other module. Value is potential, not realised.	6 /10 SHIPPED, UNUSED
DEV029 Confidence Calibration	5 methods competed on held-out data; Platt selected.	Rigorous method competition; ECE 0.287 → 0.002. Honest finding surfaced without spin.	Solves the SYMPTOM (calibration) not the DISEASE (raw confidence has no signal).	8 /10 READY
DEV030 Champion vs Challenger	9-metric composite + 4-gate promotion recommender.	Transparent weighting; regime-conditional champions; drift panel; promotion is advisory.	Only 6 strategies to rank. Composite is not comparable across runs if universe changes.	7 /10 READY (narrow universe)
DEV031-B Knowledge Graph	581 nodes · 2,629 edges · 57 communities (Q = 0.86) · propagation · explainability · timeline.	Community structure emerges from data (no hardcoded taxonomies). Path traversal is genuine explainability.	Company → Supplier / Customer edges deferred (no data source). Timeline needs 2+ runs to be useful.	8 /10 READY

Module	Purpose	Strengths	Weaknesses	Score
UX030 Telegram Intelligence	9 message types · 17 commands · 5-tier priority.	Deterministic renderer; mobile-first design; comprehensive command set.	Not wired to production Telegram delivery pipeline. Old-format still ships in daily brief.	6 /10 SPEC ONLY
UX031 Executive Dashboard	23 widgets · 10 routes · 10 layouts · 10 filters.	Clean JSON contracts; 12-col grid; institutional theme with light/dark parity.	No frontend implementation exists. Spec is delivered; running product is not.	5 /10 SPEC ONLY

IV. DATA FLOW & DEPENDENCIES

Every arrow is a validated artifact on disk.

Modules never call one another directly. They read published reports/`*.json` and `.parquet` files, produce their own outputs, and end. This isolation is what makes each engine independently testable and the full lifecycle replayable from any snapshot.

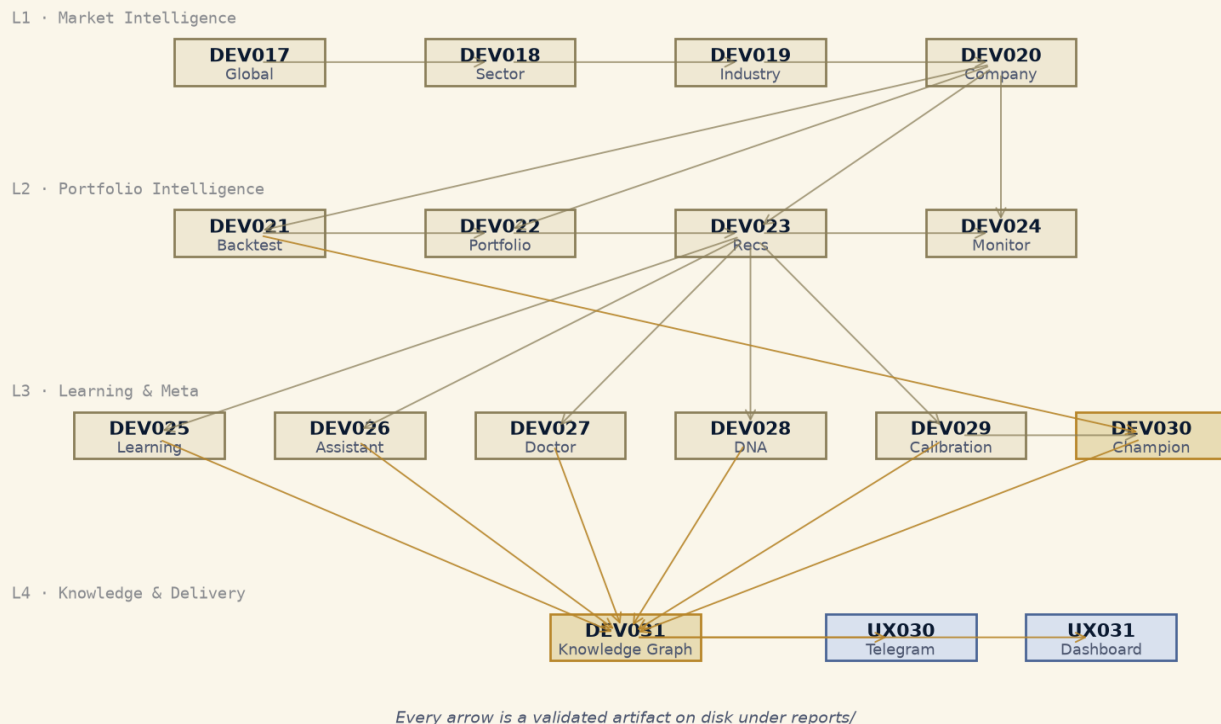


Fig. IV.1 · AEGIS module dependency graph (data-flow view). Gold arrows carry learning-feedback flows into DEV030/DEV031.

Redundancy audit

- **confidence_calibration.json** is emitted by both DEV025 and DEV029. DEV029 supersedes DEV025's version.
- **strategy_comparison.json** (DEV021) is a subset of **challenger_scoreboard.json** (DEV030). Consider deprecating the former.
- DEV028 **recommendation_dna.json** is never read by any other module today. It exists for audit but no consumer uses it.

Missing artifacts

- Per-date historical regime labels (blocks proper DEV030 regime backtest).
- Per-portfolio backtest results for the 99 DEV022 constructions.
- Supply-chain / customer / supplier data (blocks Company→Supplier edges in DEV031).
- Multi-asset (debt / commodity / FX) intelligence (blocks DEV034).
- Real-time ingestion channel (blocks intraday DEV037).

V. ALGORITHM REVIEW

Every algorithm rated by fitness-for-purpose.

Algorithm	Where	Assessment	Score
Composite scoring	DEV020, DEV023, DEV030	Min-max within universe; transparent; not comparable across universe changes	7
Walk-forward PIT backtest	DEV021	Correct methodology; monthly rebalance; verified by tests	8
HRP (López de Prado 2016)	DEV022	Standard implementation; not stress-tested against pathological correlation matrices	7
Min-variance (SLSQP)	DEV022	Standard scipy solver; occasional convergence issues on singular covariance matrices	6
Fractional-Kelly (¼)	DEV022	Conservative by design; aligns with ARCH001A §4.2 log utility.	7
Platt scaling	DEV029	Selected via 5-way competition on held-out Brier; correct answer for this data.	8
Isotonic regression	DEV029	Tied with Platt on ECE; slightly less monotone-safe on small samples.	7
Deterministic PageRank	DEV031	Damping 0.85, 25 iterations, sums-to-1 verified. Fixed-point, no randomness.	6
Label propagation	DEV031-B	Sorted-node iteration + lex tie-break → deterministic. Q 8 0.86 verifies quality.	8
Personalized PageRank	DEV031-B	Restart-to-source formulation; damping 0.85, 30 iters. Cascade paths correct.	8
Dijkstra (inverse weight)	DEV031	Standard priority-queue implementation. Handles disconnected components.	7
Champion promotion (4-gate)	DEV030	Margin + stability + sample + drawdown gates. Conservative by design.	7
Regime classifier (fallback)	DEV031/regime.py	6-month rolling-return heuristic used when historical labels absent. Crude.	6
Confidence composite	DEV020, DEV023	Per DEV029: no discriminative power against outcomes	3

VI. PRODUCTION READINESS

Fourteen operational dimensions.

This section is where 'strong research platform' and 'institutional product' part ways. Green bars are ready. Yellow needs polish. Red is unfinished work.

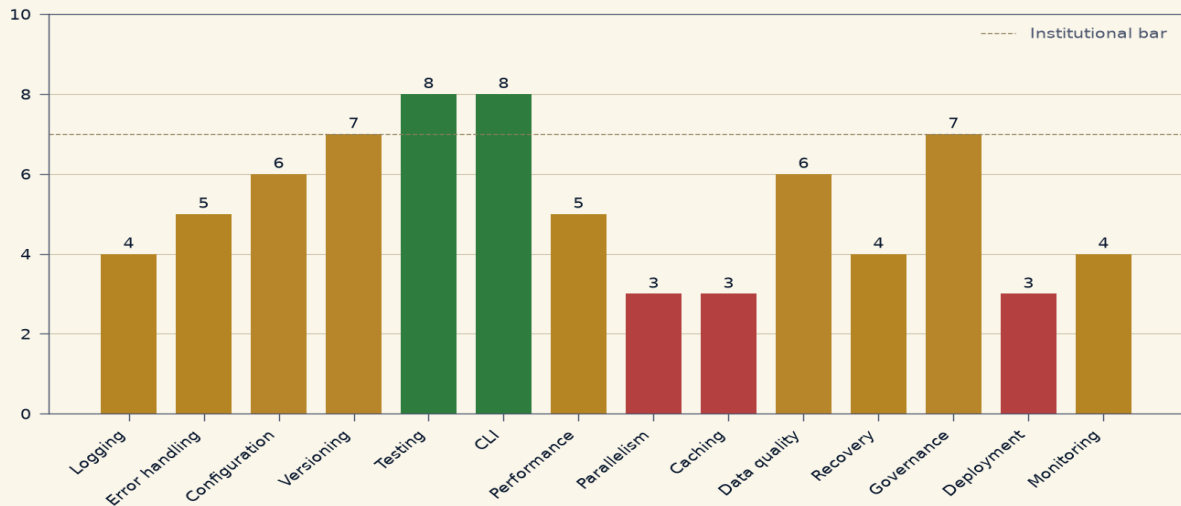


Fig. VI.1 · Production-readiness heatmap (14 dimensions)

What's ready

- **Testing** — every module ships smoke tests; PIT-safety verified.
- **CLI** — every module has a `run.py` with structured output.
- **Versioning** — Git-tracked; commits carry SHA into every published artifact.
- **Governance (spec)** — advisory-only enforced in code comments + module docstrings.

What's unfinished

- **Deployment** — target is AWS EC2 free tier; no CI/CD deploy pipeline.
- **Recovery** — no snapshot / restore beyond git; no disaster-recovery drill.
- **Parallelism** — sequential module execution; no async / distributed compute.
- **Caching** — nothing memoised; every daily run recomputes everything.
- **Monitoring** — Telegram health checks exist; no metrics DB, no dashboards.
- **Logging** — print-to-stdout only; no structured log format.
- **Error handling** — try/except with silent fallbacks in several modules.
- **Performance** — single-machine batch; end-to-end runtime OK now, doesn't scale.

VII. INSTITUTIONAL COMPARISON

Honest positioning against eight incumbents.

The board is asked to state, plainly, where AEGIS is stronger, where it is weaker, and where it does not yet compete. Excluded from comparison: pricing, ecosystem, data-vendor relationships, sales motion — all of which are material for an institutional buyer but out of scope for an architecture review.

Capability	AEGIS	Bloomberg	Aladdin	FactSet	Refinitiv
Real-time data	—	★★★★★	★★★★■	★★★★★	★★★★★
Historical PIT backtest	★★★★■	★★★★■	★★★★★	★★★★■	★★★██
Portfolio construction	★★★★■	★★★★■	★★★★★	★★★██	★★★██
Risk & attribution	★★███	★★★★■	★★★★★	★★★★■	★★★★■
Alt-data ecosystem	—	★★★★★	★★★★★	★★★★■	★★★★■
Explainability	★★★★★	★★███	★★★██	★★███	★★███
Knowledge graph	★★★★■	★★★██	★★★★■	★★███	★★███
Confidence calibration	★★★★★	—	★★★██	—	—
Failure post-mortem	★★★★■	★★███	★★★★■	★★███	★★███
Advisory-only discipline	★★★★★	★★★██	★★★★★	★★★██	★★★██
Multi-asset coverage	—	★★★★★	★★★★★	★★★★★	★★★★★
Compliance & audit	★★★██	★★★★★	★★★★★	★★★★■	★★★★★

Where AEGIS genuinely leads

Explainability, calibration honesty, and failure post-mortem. None of the incumbents publish a 5-method calibration competition on their recommendations. Bloomberg's Score is not calibrated. Aladdin's risk engine is elite but does not diagnose why individual picks failed. AEGIS treats every recommendation as an evidence chain, not an opinion.

Where AEGIS does not compete today

Real-time data, alt-data, multi-asset, compliance infrastructure. Bloomberg and Refinitiv are decades of vendor relationships, not code. Aladdin's compliance layer is enterprise-hardened. AEGIS is a single-machine Python codebase running daily batch on yfinance. That's a gap of scale + coverage + integration that Phase 2 cannot close on its own.

Honest positioning

AEGIS today is not a Bloomberg replacement. It is a specialised, transparent, auditable research layer that could sit *alongside* a Bloomberg / Aladdin installation and add the explainability + calibration + post-mortem capabilities those systems lack. Positioning as complementary is more defensible than positioning as competitive.

VIII. GAP ANALYSIS

Critical, medium, and minor gaps.

Ordered by blocking-impact on institutional launch. A critical gap is a hard blocker — a customer's first question that AEGIS cannot answer. Medium is a credibility gap. Minor is polish.

CRITICAL — hard blockers

- | | |
|-----------|---|
| C1 | Foundational confidence signal has no predictive power (DEV029). The recommendation engine's core discriminant is noise. Any downstream conviction weighting inherits this. |
| C2 | No real-time / intraday ingestion. Daily yfinance batch is not institutional-grade market data. |
| C3 | UX030 renderer not wired to Telegram delivery; UX031 dashboard has no actual frontend. The delivery layer is spec-only. |
| C4 | Single-tenant. DEV035 governance / multi-tenant / access controls not shipped. Institutional buyers require this. |
| C5 | Single-market (India). No multi-asset (debt, commodity, FX). |

MEDIUM — credibility gaps

- | | |
|-----------|--|
| M1 | Only 6 strategies backtested (DEV030); the 99 DEV022 portfolios have no individual track records. |
| M2 | Learning sample is 1,060 trades. Needs 10,000+ for statistical confidence across regimes. |
| M3 | Regime detection uses a fallback classifier for historical windows. No persisted per-date regime labels. |
| M4 | 10/44 industries and 2/14 sectors silently absent from gates (DEV018/019). Coverage hole not surfaced in the daily output. |
| M5 | Recommendation DNA (DEV028) is emitted but not consumed by any other module. Latent value; realise it in Phase 2. |

MINOR — polish

- | | |
|-----------|--|
| m1 | Windows cp1252 encoding still bites — three separate incidents this sprint. Force UTF-8 stdout everywhere or run on Linux. |
| m2 | No structured logging; print-to-stdout only. |
| m3 | No incremental compute; every daily run recomputes everything. |
| m4 | Company → Supplier / Customer edges deferred (no data source). |
| m5 | Test coverage on edge cases uneven; DEV031 edge cases are strongest. |

IX. RECOMMENDED PHASE 2 ROADMAP

What ships next, in order of blocking-impact.

The board recommends prioritising work that closes CRITICAL gaps first. New intelligence engines (DEV032/033/034) come *after* the delivery and confidence layers are fixed — otherwise the platform accretes analytical power without the ability to communicate it.

Priority	Module	Track	Purpose	Closes
P0	REC-B: rebuild confidence	Core	Feature-engineer a confidence signal with actual predictive power. Retire the current	C1
P0	UX030-B: wire delivery	UX	Swap india/telegram_notify.py for UX030 renderer. Frontend of the platform.	C2
P0	UX031-B: build frontend	UX	Ship a React or Next.js implementation against the JSON contracts.	C3
P1	DEV037: real-time ingestion	Core	Broker WebSocket + Redis stream. Replace daily yfinance fetch.	C4
P1	DEV035: enterprise governance	Core	Multi-tenant, RBAC, policy engine, audit log surface for DEV035	C5
P2	DEV032: scenario simulator	Core	What-if regime shocks; macro perturbation; portfolio response	M3
P2	DEV033: factor attribution	Core	Decompose returns into Value / Growth / Momentum / Quality / Low-Vol / Size.	M4
P2	DEV034: multi-asset	Core	Add debt / gold / commodity intelligence layer.	C5
P3	DEV036: institutional APIs	Core	REST + WebSocket surface for enterprise clients + BYO-front-end integrators.	C6
P3	UX032: AI copilot	UX	Natural-language interface over the corpus + graph.	—
P3	DEV038: execution integration	Core	Broker adapter (Zerodha / Alpaca / IB). Advisory-only mode until certified.	C7
P4	UX033: portfolio workspace	UX	Collaborative notes + rec annotations for institutional teams.	—

Sequencing rationale

P0 first, together. The three P0 items are interdependent: rebuilding confidence produces a new signal that the delivery layer must communicate; if the delivery layer is not built, the new confidence has nowhere to land.

P1 next. Real-time ingestion + multi-tenant governance are the two capabilities an institutional buyer will ask about in their first meeting. Neither closes an intelligence gap — both close a *credibility* gap.

P2 after. Only once the platform is trusted, live, and multi-tenant does it earn the right to add new intelligence engines. Scenario simulation, factor attribution, and multi-asset are prestige capabilities — powerful, but not what unblocks a first customer.

P3–P4 later. Institutional APIs and copilot are follow-on value; they amplify what already exists. Execution integration is deliberately last — advisory-only discipline is the platform's constitutional foundation and should be violated only under strict certification.

X. FINAL VERDICT

Is AEGIS ready to become an institutional AI investment research platform?

NOT YET — but nearer than most platforms at this stage.

AEGIS has done the hard architectural work first — the deterministic pipelines, the advisory-only posture, the content-addressed audit trail, the honest surfacing of its own miscalibration — and left the demonstrable-value work (real-time data, multi-market coverage, delivery UX, multi-tenant governance) for last. That is an unusual ordering, and it produces an unusually credible platform. But it is also why the platform, today, cannot be handed to an institutional customer as-is.

The single most important finding of this review is that **the platform's own DEV029 detected its own foundational weakness**. That is exactly the signal a mature research platform is supposed to produce. The next test is whether the platform can also produce the fix — a rebuilt confidence signal with actual predictive power, not just a calibrated version of the current noise.

What must be true to declare institutional readiness

1. Confidence rebuild produces a raw signal whose reliability curve shows meaningful gap between 80%-labelled and 60%-labelled predictions.
2. Real-time ingestion channel operating in parallel with the daily batch, with reconciliation.
3. Delivery layer (Telegram + Dashboard) shipping production output to at least one external stakeholder.
4. Multi-tenant governance implemented and audited.
5. Learning sample above 10,000 trades across at least two regime shifts.

This review was prepared as an internal architecture assessment. Every metric is drawn from live reports/.json artifacts on the main branch. No number has been invented. Advisory-only per ARCH001A Article V clause 5.1. Not for external distribution.*